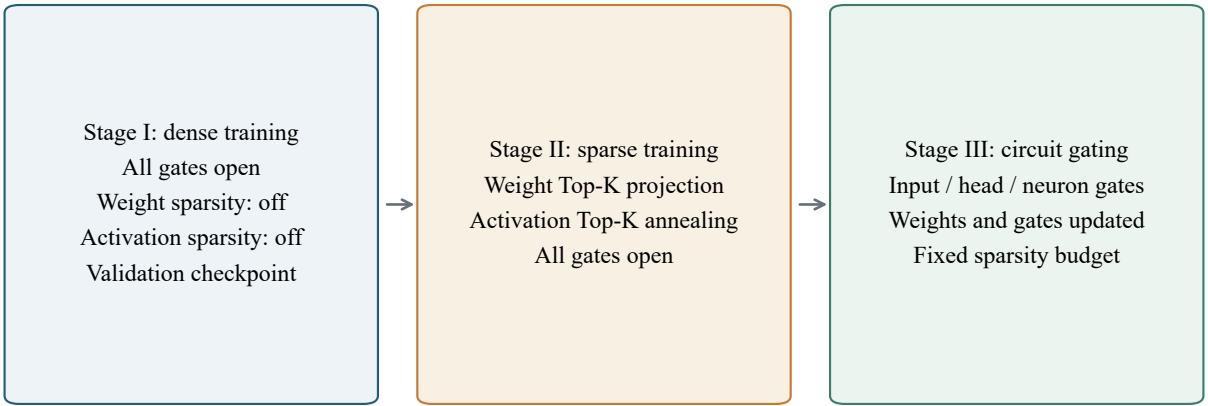

Projection is reapplied after every sparse optimizer update.
Each phase restores its best validation checkpoint.